

Jialin Zhang

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Education

Tongji University, Computer Science, GPA: 4.40/5.00 (89.02/100) 2023-Present

Relevant courses: Mathematical Analysis, General Physics, Linear Algebra, CPP, Data Structures, Discrete Mathematics, etc.

Project Experience

Research in Machine Learning, Deep Learning and Diffusion model

- AI: Applications of ML and DL. -Taught by Björn Schuller (IC) 2024.09-2024.11
 - Learn the basic principles of DL and ML, Grade: 86/100
- AI and ML, HKU Summer School. -Advised by Yi Ma (HKU) 2024.07-2024.08
 - Creation and Basic Application of Diffusion Models. -Details: GitHub Link

Research in LLMs, KGE(Knowledge Graph Embedding)

- Salary Management System Based on LLM and KGE 2024.11-Present
 - Build job network based on Neo4j, Use KGE models to find potential relations

Research in Computer Vision (CV)

- Underwater image enhancement (2024APMCM, Third Price) 2024.11.21-25
 - Traditional methods: *White Balance*, *Histogram Equalization*, *Retinex*, *Dehazing*.
 - Attempted DL methods based on CNN, but lose due to lack of a training dataset.
- Suspended Sediment Concentration in Yangtze Estuary. 2024.02-Present
 - PyTorch + DL model for image preprocessing tasks.
- Automated Bridge Health Monitoring Using Infrared Technology 2023.09-2024.01
 - OpenCV to classify infrared images and DL model to identify abnormal points
 - Python for shortest path planning based on DFS algorithm. -Details: GitHub Link
- Tooth root canal orifice detection (CPP + Pytorch) 2024.10-2024.11
 - VGG for image classification and DL model for image segmentation

Website Deployment Training

- Build and maintain a mental-health website 2024.10-2024.12
 - Locally deploy website using the PyCharm+Django+MySQL/SQLite framework.

Additional Training:

- Entertainment Industry Introduction. Taught by Mark Young (USC) 2024.02-2024.06
 - Learned the operational mechanisms of the film industry. Grade:Passed (P/F)

Honors

Tongji University Freshman Scholarship	2023
Tongji University Outstanding Undergraduate Scholarship	2024
Social Activity Scholarship of Tongji University for the Academic	2023-2024
Honorable Mention, 2025 MCM	2025
Third Prize, 2024 APMCM	2024
Academic Year Outstanding Mental Health Counselor at Tongji U	2023-2024

Others

Tongji U Chess Club - Head of the Chess Department	2024.06 - Present
Outstanding Member of Guohao College, Tongji University	2023.09 - 2024.07